

RESEARCH LETTER

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Key Points:

- A degree 900 gravity field from the GRAIL Primary and Extended Mission
- The GRAIL Extended Mission doubles the resolution of the gravity field
- The Bouguer spectrum is well determined to near harmonic degree 550

Correspondence to:

A. S. Konopliv,
Alexander.S.Konopliv@jpl.nasa.gov

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High-resolution lunar gravity fields from the GRAIL Primary and Extended Missions

Alex S. Konopliv¹, Ryan S. Park¹, Dah-Ning Yuan¹, Sami W. Asmar¹, Michael M. Watkins¹, James G. Williams¹, Eugene Fahnestock¹, Gerhard Kruizinga¹, Meegyeong Paik¹, Dmitry Strekalov¹, Nate Harvey¹, David E. Smith², and Maria T. Zuber²
¹Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California, USA, ²Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, Massachusetts, USA

Abstract The resolution and accuracy of the lunar spherical harmonic gravity field have been dramatically improved as a result of the Gravity Recovery and Interior Laboratory (GRAIL) mission. From the Primary Mission, previous harmonic gravity fields resulted in an average $n = 420$ surface resolution and a Bouguer spectrum to $n = 330$. The GRAIL Extended Mission improves the resolution due to a lower average 23 km altitude orbit. As a result, new harmonic degree 900 gravity fields (GL0900C and GL0900D) show nearly a factor of 2 improvement with an average surface resolution $n = 870$ and the Bouguer spectrum extended to $n = 550$. Since the minimum spacecraft altitude varies spatially between 3 km and 23 km, the surface resolution is variable from near $n = 680$ for the central farside to near $n = 900$ for the polar regions. These gravity fields with 0.8 million parameters are by far the highest-degree fields of any planet ever estimated with a fully dynamic least squares technique using spacecraft tracking data.

1. Introduction

Prior to the Gravity Recovery and Interior Laboratory (GRAIL) mission [Zuber *et al.*, 2013a], multiple missions contributed to the determination of the lunar gravity field including Lunar Orbiter I–V, Apollo 15 and 16 subsatellites, Clementine [Zuber *et al.*, 1994], Lunar Prospector (LP) [Konopliv *et al.*, 2001; Mazarico *et al.*, 2010], and SELENE [Goossens *et al.*, 2011]. However, LP predominately determined the nearside gravity field to near harmonic degree $n = 150$ and SELENE, with the first direct measurement of the lunar farside, improved the farside resolution to near $n = 70$. Compared to the historical Deep Space Network (DSN) S band tracking data, the collection of the 4 orders of magnitude more accurate GRAIL interspacecraft tracking data from both the Primary and Extended Missions has significantly superseded contributions to the static gravity field from all previous missions. The two GRAIL spacecraft (Ebb, GRAIL-A, and Flow, GRAIL-B) measurement system was based upon the terrestrial Gravity Recovery and Climate Experiment (GRACE) mission [Tapley *et al.*, 2004]. GRAIL measurements are from a dual Ka band ranging system [Klipstein *et al.*, 2013] resulting in an interspacecraft range rate accuracy of $0.03 \mu\text{m/s}$ with 5 s samples for the 89 d Primary Mission and $0.05 \mu\text{m/s}$ with 2 s samples for the 106 d Extended Mission. The GRAIL measurement noise is noticeably lower than the GRACE noise ($\sim 0.2 \mu\text{m/s}$ 5 s samples) since there are no dual-frequency corrections for the ionosphere, and the link performance is improved.

The lunar gravity field derived from the GRAIL Primary Mission [Zuber *et al.*, 2013b; Konopliv *et al.*, 2013; Lemoine *et al.*, 2013] has shown strong global 98% coherence with topography for all well-determined degrees higher than $n = 120$ up to about $n = 330$ and local regions to $n = 550$ [Han, 2013]. The average surface resolution, defined as where the unconstrained gravity coefficient root-mean-square (RMS) error intersects the expected signal, is $n = 420$ or a half-wavelength surface resolution of 13 km. After subtraction of the gravity signal expected from Lunar Reconnaissance Orbiter (LRO) topography [Smith *et al.*, 2010], the corresponding Bouguer gravity to $n = 330$ indicates a lighter than expected lunar crustal density and thus higher porosity [Wieczorek *et al.*, 2013], and the Bouguer gravity gradient displays linear anomalies that are possible subsurface dikes representing evidence of early lunar expansion [Andrews-Hanna *et al.*, 2013]. The improved long-wavelength spectrum of the gravity, in particular $n = 2$ and 3, together with lunar laser ranging measurements, has led to a factor of 5 improvement in the lunar moments of inertia and second-degree elastic tidal Love number k_2 as well as the first detection of the k_3 [Konopliv *et al.*, 2013; Lemoine *et al.*, 2013]. In addition to studies of the internal structure of the Moon, the improved gravity field of the Moon has significantly reduced the spacecraft orbit uncertainty and long-term prediction error for any future navigation of lunar missions.

Table 1. Summary of the Tracking Data Used in the GRAIL Primary Mission^a

	03/01/2012 16:28:00 04/03/2012 13:13:00 (32.71 Days)	04/03/2012 13:24:00 05/03/2012 00:58:00 (29.44 Days)	05/03/2012 01:09:00 05/29/2012 16:36:00 (26.55 Days)	Total
From/To Arc Length				
KBRR	561939	504234	457584	1523757
Two-way S band Doppler	182888	133976	117787	434651
One-way X band Doppler	237598	181993	226317	645908

^aThe spacecraft measurements are separated into three intervals with about one month of data each. The first (~March) and third (~May) months contain all the data arcs where part of the spacecraft orbit is in shadow. The spacecraft is in continuous full Sun from 2 April 2012 at 12:20 to 3 May 2012 at 18:41.

As with the Primary Mission data analyses, the GRAIL Extended Mission gravity results are presented by two independent but collaborative groups at the Jet Propulsion Laboratory (JPL, this paper) and Goddard Space Flight Center. The results are very similar (e.g., both extend the global 98% coherence to nearly the same harmonic degree) and demonstrate the integrity of the solutions. In addition, the mutual exchange of ideas has significantly benefited the development of software and techniques for this numerically very challenging problem. Multiple solutions (GL0900C and GL0900D) of $n = 900$ are presented here that include 0.8 million parameters estimated with a least squares filter, by far the largest spacecraft-derived gravity field to date.

2. Gravity Data and Mission Characteristics

The 3 month GRAIL Primary Mission from 1 March to 29 May 2012 collected global gravity data with an average 55 km altitude but with minimum altitudes over the surface varying between 18 and 52 km. The spacecraft separation varied linearly from an initial 80 km to 220 km near 1 April, and then drifted back linearly to a final 80 km. The data were processed in time intervals or arcs of 2–3 days that were mostly separated by spacecraft maneuvers to despin the reaction wheels used for attitude control. Each orbital arc contained continuous 5 s sample Ka band range rate (KBRR) interspacecraft data and multiple DSN passes of two-way S band data [Moyer, 2003] to help with the absolute orbit positioning for GRAIL. Tables 1 and 2 give a summary of all GRAIL data used in the solution.

The 3.5 month GRAIL Extended Mission from 30 August to 14 December 2012 collected gravity data from significantly lower orbits than the Primary Mission: an average 23 km altitude from the start to 19 November, then a 20 km altitude to 6 December, and subsequently an average altitude near 11 km up to the end of mission. Figure 1 shows the resulting minimum spacecraft altitude above the lunar surface relative to the LRO topography [Smith *et al.*, 2010]. The minimum altitudes were ~3 km near 300°E just before a low-altitude sweep over the Orientale Basin, and the higher minimum altitude regions are ~23 km near the central nearside. Since the altitudes were much lower than the Primary Mission, the spacecraft separation between the two spacecraft was also much closer (between 30 and 75 km, but typically 60 km) to avoid any multipath errors in the reception of the interspacecraft range.

The generation of the Ka band range-rate observable is a multistep process [Harvey *et al.*, 2012] that is briefly outlined in Konopliv *et al.* [2013] that requires precise orbit determination, calibration of Ka band antenna boresights, relative clock corrections between spacecraft, absolute time adjustments relative to the DSN clocks, and relativity adjustments to the observable. For the Primary Mission the range-rate data were obtained from the range data using a low-pass digital filter [Thomas, 1999] with a 70.7 s window resulting in

Table 2. Summary of the Tracking Data Used in the Extended Mission^a

	08/30/2012 16:20:00 10/10/2012 23:55:00 (41.32 Days)	10/11/2012 00:06:00 10/27/2012 02:49:00 (16.11 Days)	10/27/2012 03:00:00 12/14/2012 20:56:00 (48.75 Days)	Total
From/To Arc Length				
KBRR	1763778	690047	2082798	4536623
Two-way S band Doppler	217870	71214	247177	536261
One-way X band Doppler	355759	114758	378885	849402

^aThe spacecraft measurements are separated into three intervals with about 1 month of data each. The spacecraft is in continuous full Sun from 11 October 2012 at 02:32 to 28 October 2012 at 07:58.

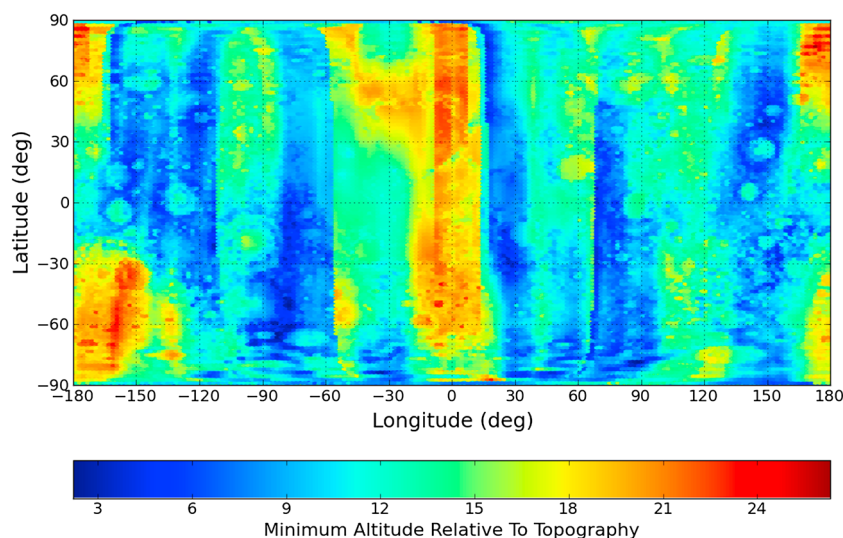


Figure 1. Minimum spacecraft altitude (in km) over the actual surface topography of the Moon [Smith et al., 2010] during the Primary and Extended Missions.

5 s samples with frequencies cut off beyond harmonic degree 720. The Extended Mission filter used a 74.7 s window giving 2 s samples with frequencies included to $n = 1800$.

3. Gravity Estimation Technique

The least squares technique for determining the lunar gravity field from GRAIL data is outlined in detail in the presentation of the Primary Mission results [Konopliv et al., 2013] and the description of the GRAIL measurement system [Asmar et al., 2013]. Additionally, the procedure is similar to the lower resolution GRACE gravity studies for the Earth [Tapley et al., 2004]. For this study, the number of parameters (812,000) in the spherical harmonic degree $n = 900$ field is about double those of the Primary Mission solution GL0660B [Konopliv et al., 2013] and requires about 4 times the computational resources. For the most recent solution, the runtime required on the NASA Ames Pleiades Supercomputer was ~ 200 h using 200 Ivy Bridge nodes with 20 cores each. The least squares filter dominates the computation requirements ($\sim 95\%$), whereas the generation of the observation equations (numerical integration of the spacecraft orbit and processing KBRR and DSN data) required only a small fraction of the computer time ($< 5\%$). The filter is based upon QR factorization (Q is an orthogonal matrix and R is an upper triangular matrix) using the publicly available ScaLAPACK software package that is implemented using MPI (Message Passing Interface; see <http://www.netlib.org/scalapack>). The Primary and Extended Mission observations were processed into one and two upper triangular square root information arrays (or SRIF arrays), respectively. Each SRIF array (2.6 TB for $n = 900$) was stored in memory across the 200 compute nodes.

4. Gravity Results

Two gravity solutions to harmonic degree 900 were determined using the entire GRAIL data set. The solution GL0900C represents the first JPL solution complete to degree 900 that was a significant improvement over the nominal gravity field to $n = 780$. An important consideration in the solution strategy is the data accuracy assigned to the observables. For the first solution, we used a KBRR uncertainty between 0.03 and 0.06 $\mu\text{m/s}$ for the Primary Mission, with the higher uncertainty for orbits near the beginning of March and end of May due to nongravitational errors from extended shadowing of the spacecraft and higher thermal accelerations. The Extended Mission used a uniform 0.1 $\mu\text{m/s}$ accuracy except for the lower altitude orbits beginning 19 November, which used a typical 0.5 $\mu\text{m/s}$, higher than the raw noise to avoid aliasing the gravity solution due to large systematic signatures in gravity from degrees higher than 900.

The second solution, GL0900D, used GL0900C as the nominal solution and represents a complete iteration of the gravity field coefficients to $n = 900$ (i.e., using local converged arcs with GL0900C as the starting point, we estimate new corrections to the GL0900C field). The 25° radius north and south polar cap regions are shown

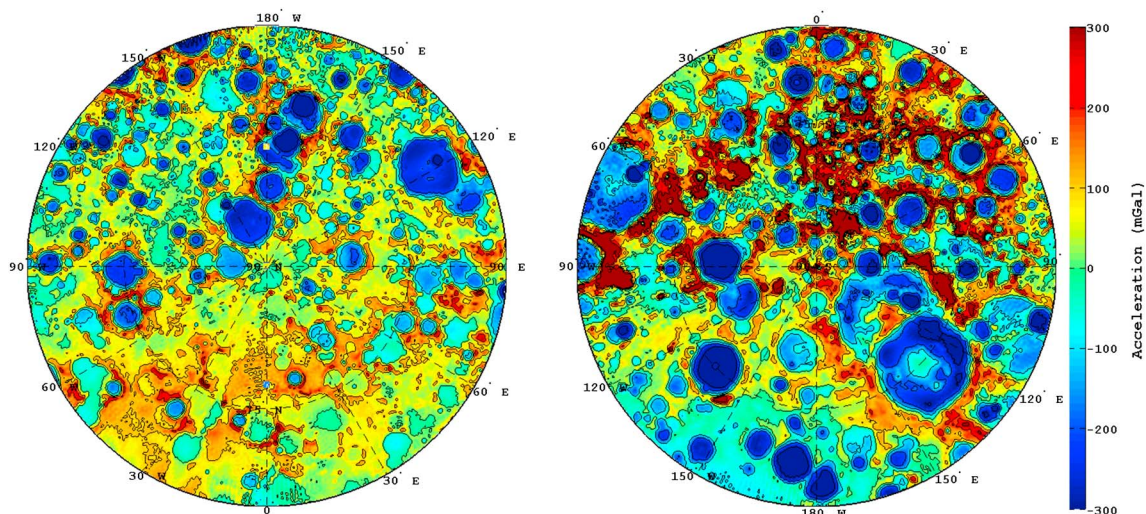


Figure 2. High-resolution image of the (left) northern and (right) southern polar regions of the Moon showing the vertical acceleration on the 1738 km reference sphere for the GL0900D gravity field truncated at degree 800 for latitudes beyond 65°. The projection is polar stereographic. Maximum and minimum values for the northern and southern polar regions are 409, −449 and 1269, −475 mGals, respectively.

for GL0900D in Figure 2 to demonstrate the high-resolution detail of craters resolved by the gravity field. The RMS difference over these regions between the two solutions is 2.4 mGals with a peak difference of 30 mGals. This is mostly in agreement with the formal error from the covariance (RMS over the region of 3.0 mGals and 11 mGal maximum). The weighting scheme used for the GL0900D solution was adjusted based upon what we learned from the postfit KBRR RMS fit of all arcs using the GL0900C solution. The weighting remained the

same for the Primary Mission data, but the uncertainties for the Extended Mission were reduced by about a factor of 2 to 0.05 $\mu\text{m/s}$ for data arcs up to 19 November and up to 0.2 $\mu\text{m/s}$ thereafter depending on postfit RMS residuals of the KBRR data. We additionally reduced the weight for arcs after 19 November in attempt to extract more gravity information but also limit aliasing.

Another difference between the two $n = 900$ solutions is the a priori constraint applied to smooth the coefficients over regional areas where the solution is not as well determined. The GL0900C solution uses a Kaula power law constraint of $2.5 \times 10^{-4}/n^2$ that biases the coefficients toward zero with an a priori uncertainty based upon the harmonic degree, but it is only applied for degrees $n > 600$. Although the Kaula power law matches most of the gravity spectrum, greater power is expected at higher degrees based upon the Lunar Orbiter Laser Altimeter (LOLA) topography [cf. Smith et al., 2010]

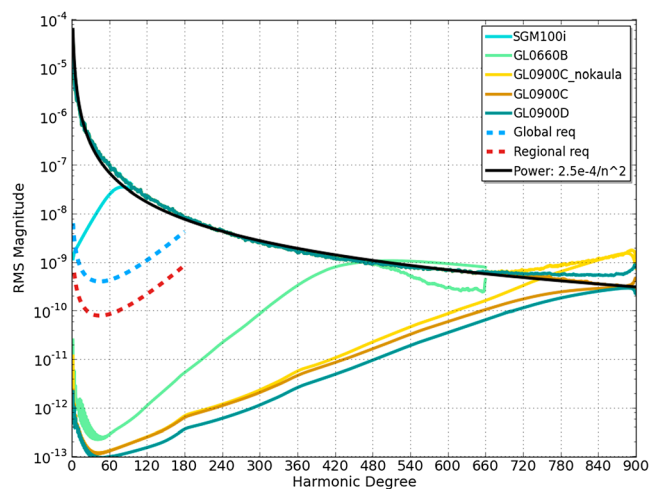


Figure 3. The full gravity spectrum for the best pre-GRAIL gravity error spectrum (SGM100i) based upon SELENE data [Goossens et al., 2011], the GRAIL Primary Mission solution GL0660B [Konopliv et al., 2013], and the two solutions (GL0900C and GL0900D) of this paper. The upper negative-sloping solid curves show the RMS normalized gravity field coefficients for each degree and the lower upward sloping solid curves show their uncertainties. All GRAIL solutions easily satisfy the mission global and regional requirements shown with dashed lines. The Kaula power law that fits the low- to medium-degree portion of the spectrum overconstrains the high degrees ($n > \sim 700$). The power of GL0900D is near that of gravity derived [e.g., Wiczorek and Phillips, 1998] from LOLA topography [Smith et al., 2010].

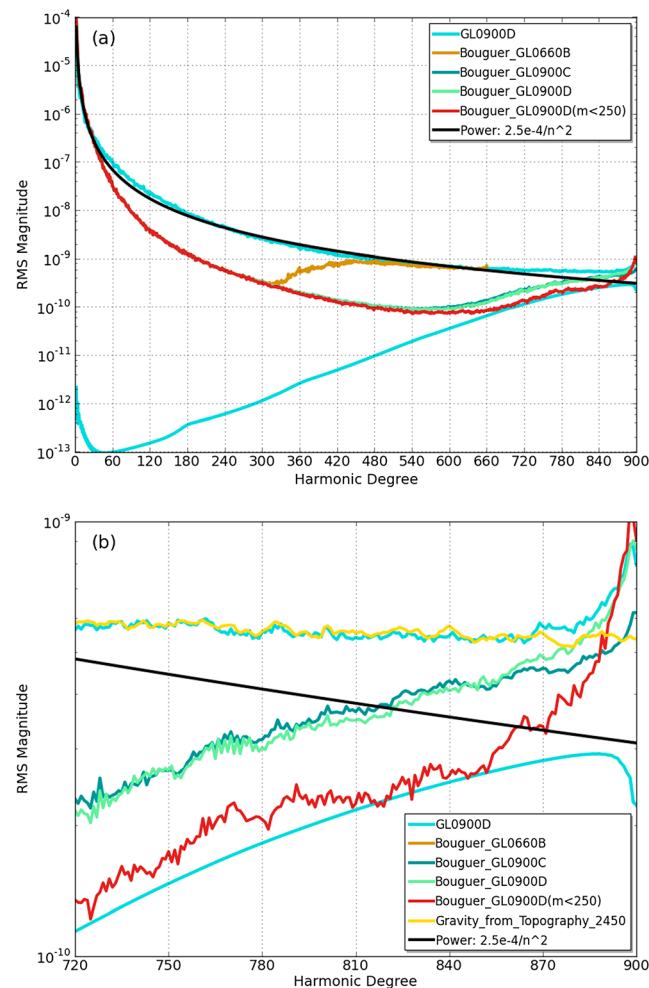


Figure 4. (a) The gravity spectrum of GL0900D and the Kaula power law are compared with the Bouguer spectrum, the difference of GRAIL-derived gravity and gravity predicted from constant density topography, for the Primary Mission gravity field GL0660B [Konopliv *et al.*, 2013] and the solutions of this paper, GL0900C and GL0900D. Additionally, the Bouguer spectrum is shown for the better-determined low-order terms of the gravity field ($m < 250$). (b) The high-degree end of the spectrum magnified shows the benefit of the low-order terms, and the close match between the GL0900D power and the power from the topography converted to gravity.

approaches the gravity signal, is near $n = 420$ for the Primary Mission gravity field, whereas the new average global resolution is about $n = 870$.

The Bouguer spectrum of Figure 4, a factor of 10 smaller in magnitude than the gravity spectrum, shows the close match between the gravity determined by GRAIL and gravity inferred from topography. The Primary Mission match to $n = 330$ is extended to near $n = 550$ with the Extended Mission results included. At $n = 600$, the slope of the gravity uncertainty curve limits the determination of the Bouguer spectrum since it intersects it at a lower degree. However, the Bouguer differences slowly increase to match the gravity spectrum near $n = 870$, the overall global resolution of the gravity field GL0900D. Also, the lower order part of the gravity spectrum (e.g., order $m < 250$) has about a factor of 2 lower uncertainties than the higher-order terms, a consequence of the interspacecraft range-rate measurement. As a result, using the low-order part the Bouguer spectrum as shown in Figure 4 can extend the Bouguer match to about $n = 650$.

The correlations of GRAIL gravity with gravity predicted from topography in Figure 5 are likewise improved by the Extended Mission with 98% coherence [Zuber *et al.*, 2013b] extended to higher degrees by about a factor

converted to gravity (expanded to the twentieth powers of topography) [e.g., Wieczorek and Phillips, 1998] assuming a constant crustal density of 2450 kg/m^3 . The density was reduced from 2550 kg/m^3 of Wieczorek *et al.* [2013], to better match the observed power in the gravity near the end of the spectrum unaffected by the choice of the a priori Kaula constraint ($\sim n = 600$). The GL0900D solution uses a power law constraint proportional to $n^{1/2}$ at higher degrees that is close to the power expected from topography and only applies that constraint for degrees $n > 700$. So the second solution has a much looser a priori constraint that closely matches the magnitude expected from topography.

The RMS gravity spectrum and errors in Figure 3 show dramatic improvement compared to pre-GRAIL solutions (the best represented by the SELENE gravity field SGM100i) [Goossens *et al.*, 2011]. There is an approximate factor of 2 improvement in resolution for the new solutions based upon the entire GRAIL data set versus the previous Primary Mission-only solution (GL0660B) [Konopliv *et al.*, 2013]. This is almost entirely due to inclusion of the lower altitude Extended Mission data, but minor improvements were also achieved with small adjustments in the weighting of the observations and gravity constraint as discussed above. The average global lunar gravity resolution, where the error spectrum

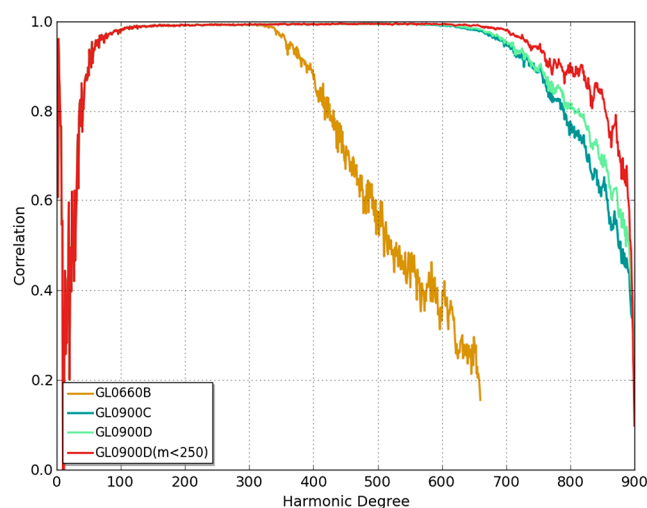


Figure 5. Global correlations of gravity fields with gravity inferred from constant density topography for the Primary Mission (GL0660B) [Konopliv *et al.*, 2013], the Primary plus Extended Missions (GL0900C and GL0900D), and the lower order part of the spectrum ($m < 250$) for GL0900D.

of 2. The best correlation is given by the GL0900D model, and its improvement is a combination of the reiteration (due to nonlinear corrections and gravity coefficients that are no longer initially zero for $n > 780$) and adjustment in data weighting. The relaxed Kaula constraint of GL0900D did not contribute to the improvement. We tested slight adjustment of the Kaula scale factor for the GL0900D solution and noted minor decrease in global correlations as the factor was increased (constraint relaxed). As with the Bouguer analysis, the correlation for the $m < 250$ part of the spectrum is increased.

The local resolution of the gravity field is determined mostly by the spacecraft altitude and by the ground track spacing, which is largest at the equator

and most dense at the poles. The local resolution, or degree strength [Konopliv *et al.*, 1999], for the Primary Mission varied between $n = 371$ and $n = 583$ [Konopliv *et al.*, 2013, Figure 13]. For solutions with the Extended Mission, we modified the technique slightly to use clone fields from the constrained covariance matrix. The original technique found the degree where the surface acceleration error per harmonic degree for each grid location on the surface equals the expected acceleration signal using a Kaula power law. The acceleration at each degree was computed using the truncated unconstrained covariance for degrees 2 to n , which is equivalent to higher-degree terms not being estimated. Here instead, we use the full constrained covariance and plot the harmonic degree where the uncertainty for a given degree is 70% of the expected signal. The resulting map in Figure 6 shows the variable surface resolution of the GL0900D field with a minimum $n = 688$ and maximum $n = 900$. The maximum resolution from the covariance is for the polar regions due to the higher concentration of data from coalescing ground tracks, and the indicated resolution may be slightly optimistic.

In addition to lunar geophysical studies, the extremely accurate GRAIL gravity field has significantly improved the orbit determination of lunar missions. Orbit errors are assessed using adjacent 2 day data arc solutions from the S band Lunar Prospector mission. Each LP spacecraft orbit arc solution is propagated by one orbit (2 h) to overlap with the next arc. The RMS differences in the orbits are then computed. Using the pre-GRAIL LP150Q [Konopliv *et al.*, 2001] field, the higher-altitude (~ 100 km) RMS overlap errors for the nominal mission

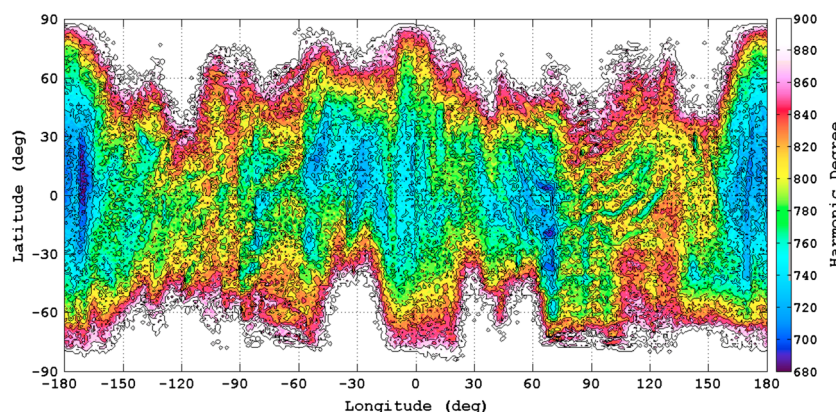


Figure 6. The resolution of gravity field GL0900D as a function of latitude and longitude from the solution covariance.

in the radial, along-track, and orbit normal directions (or RTN) are 0.3, 3.5, and 2.6 m and the lower altitude (~35 km) orbit overlap errors are 1.7, 23.8, and 18 m. With the GRAIL primary mission field GL0660B, the LP results were similar for the nominal mission but about 3 times better for the LP extended mission [Konopliv *et al.*, 2013]. With the new GL0900D gravity field, all LP gravity errors have been essentially removed. The overlap errors for the LP nominal and extended missions are now nearly equal, with a combined 0.27, 2.8, and 2.2 m for RTN errors, respectively, with the error now limited by the quality of the S band data.

Analysis of GRAIL data strongly improves the resolution and accuracy of the Moon's spherical harmonic gravity field. The resulting field exhibits unprecedented detail for basins, craters, and structures on and beneath the lunar surface.

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